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NB-IoT vs. LoRaWAN: An Experimental Evaluation for Industrial Applications

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Industrial Internet of Things (IIoT)

improve crucial factors, such as low power consumption and communication range

Following the rapid IoT market expansion, LPWAN has become one of the faster-growing areas in IoT. Low Power Wireless Area Network (LPWAN) (common term to identify the wireless technologies)

Narrow-Band Internet of Things (NB-IoT). Among them, LoRa and NB-IoT are the two leading technologies

NB-IoT combines the benefits of the 4G mobile network, namely the global coverage and the long-range, with the energy efficiency typical of LPWANs provide better indoor coverage support for a massive number of low-throughput devices

This paper evaluates LoRaWAN and NB-IoT as wireless communication technologies for industrial application scenarios that require to transfer a few bytes per day. The evaluation is based on experimental results obtained in-field expecting a sensor node for crack measurements in civil structures. We test both technologies with experimental results in different coverage conditions, intending to assess the energy consumption, the estimated battery lifetime, and the packet loss. Our assessment shows that LoRaWAN outperforms NB-IoT in terms of energy consumption. In an application where buffering is not allowed, the LoRaWAN protocol increases

the battery life up to $10\times$ against NB-IoT: for Packet 3 scenario (36 Bytes payload), DR2 / M Coverage, NB-IoT EPB is $10\times$ higher compared to LoRaWAN. However, NB-IoT is adequate for applications where information can be buffered on the node because the energy for each transmission is almost independent of the payload size. For example, in Packet 33 scenario (396 Bytes payload), DR2 / M Coverage, if high delivery latency is tolerable, NB-IoT EPB is $11\times$ lower compared to LoRaWAN, due to the larger number of messages sent by LoRaWAN. Moreover, we verify that Tactive in the NB-IoT is heavily dependent on network coverage, as it grows up to $3\times$ times passing from a "Good" (average RSSI of -80dBm) coverage to a "Bad" one (average RSSI of -130dBm). On the other hand, NB-IoT offers the highest QoS, which guarantees data delivery. This feature makes it a potential replacement to LoRaWAN in all the applications where the energy constraint is not an issue or when communication reliability is a crucial factor and good NB-IoT coverage is available.